

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 1

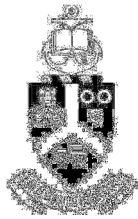
Kopiereg voorbehou

Module EBN111
Elektrisiteit en Elektronika
13 Maart 2014

Semester Test 1

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Module EBN111
Electricity and Electronics
13 March 2014



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	45	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute <i>90 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		16 + Answer sheet	

BELANGRIK- IMPORTANT

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Antwoorde moet eenhede insluit!
Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1 JD le Roux
	2 D Bhatt

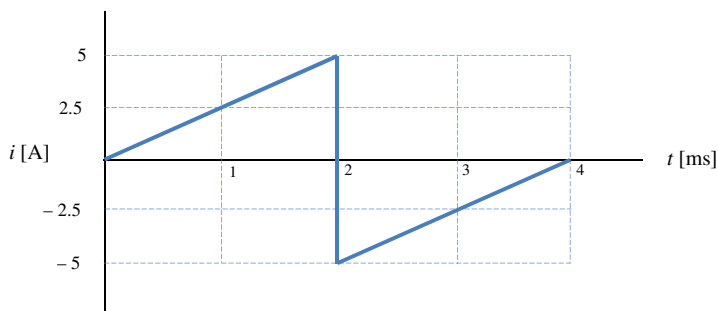
VRAAG/QUESTION 1 [7]

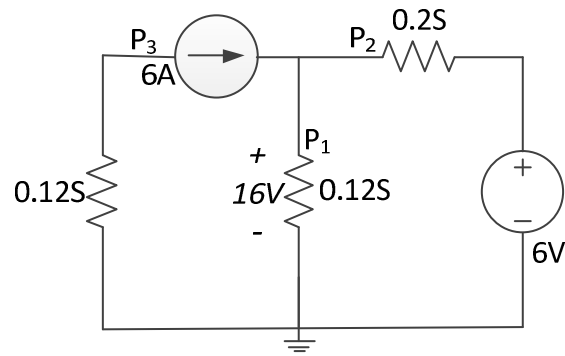
Vraag 1.1 [4]

- a) 'n LED liggie word gekoppel aan 'n 9 V battery en absorbeer energie volgens die volgende vergelyking: $w = 10(1 - 0.2t) \text{ J}$. Bepaal die drywing verbruik by $t = 5 \text{ s}$.
- b) 'n Lithium-battery voorsien 'n konstante 1 mA en 5 V vir 1 volle dag aan 'n selfoon. Bepaal die hoeveelheid energy wat verbruik is.
- c) Elektrisiteit kos R5/kWh. 'n 3kW Russel-Hobbes ketel benodig 20 sekondes om 100 milliliter water te kook. Wat is die koste om 1.5 liter water te kook?
- d) Die verandering in die stroom wat deur 'n element vloei word in die onderstaande grafiek aangedui. Bepaal die totale lading wat deur die element vloei tussen 0 ms en 2 ms.

Question 1.1 [4]

- a) An LED light is connected to a 9 V battery and absorbs energy according to the following equation: $w = 10(1 - 0.2t) \text{ J}$. Determine the power used at $t = 5 \text{ s}$.
- b) A Lithium-battery supplies a constant 1 mA and 5 V for one full day to cellphone. Determine the amount of energy used.
- c) Electricity costs R5/kWh. A 3kW Russel-Hobbes kettle needs 20 seconds to boil 100 milliliter of water. What is the cost to boil 1.5 liter of water?
- d) The change in the current through an element is shown in the graph below. Determine the total charge through the element between 0 ms and 2 ms.



Vraag 1.2 [3]Bepaal P_1 , P_2 en P_3 .**Question 1.2 [3]**Determine P_1 , P_2 and P_3 .

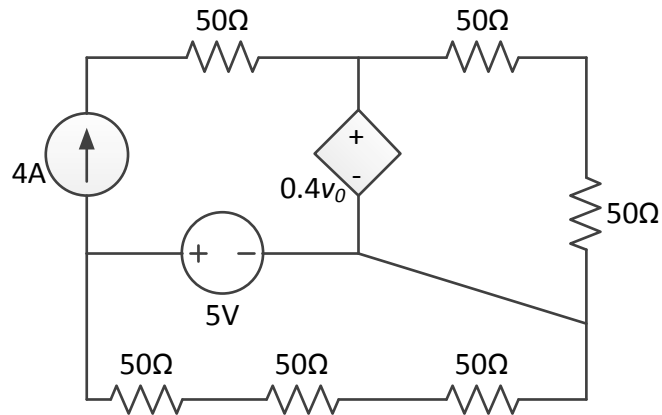
VRAAG/QUESTION 2 [16]

Vraag 2.1 [3]

Bepaal die aantal takke, nodusse en onafhanklike lusse vir onderstaande kringbaan.

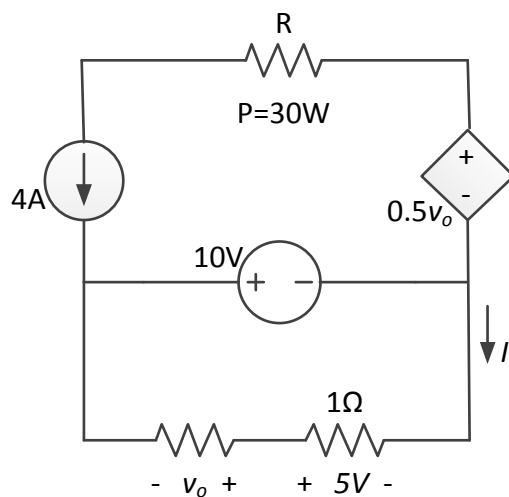
Question 2.1 [3]

Identify the number of nodes, branches and independent loops for the circuit below.



Vraag 2.2 [3]
Bepaal I , v_o , en R .

Question 2.2 [3]
Calculate I , v_o and R .

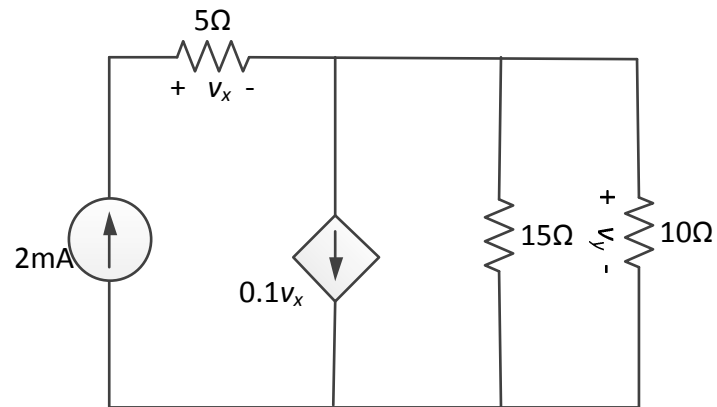


Vraag 2.3 [2]

Bepaal v_y .

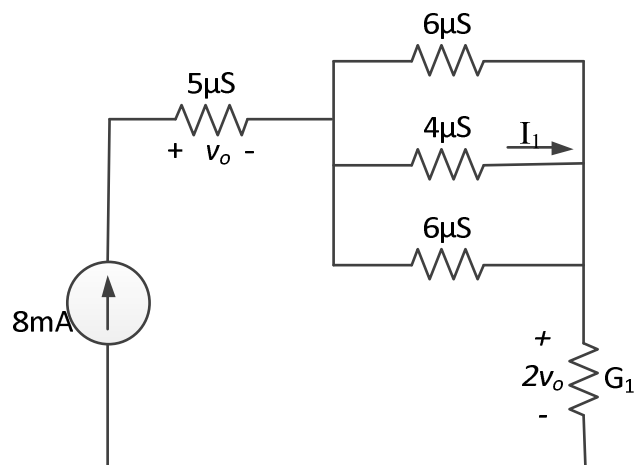
Question 2.3 [2]

Calculate v_y .



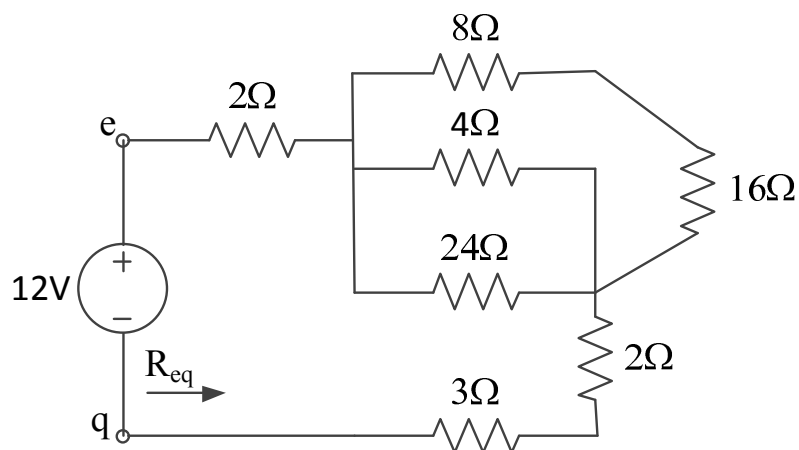
Vraag 2.4 [2]
Bepaal I_I en G_I .

Question 2.4 [2]
Calculate I_I and G_I .



Vraag 2.5 [2]
Bepaal R_{eq} .

Question 2.5 [2]
Calculate R_{eq} .



Vraag 2.6 [4]

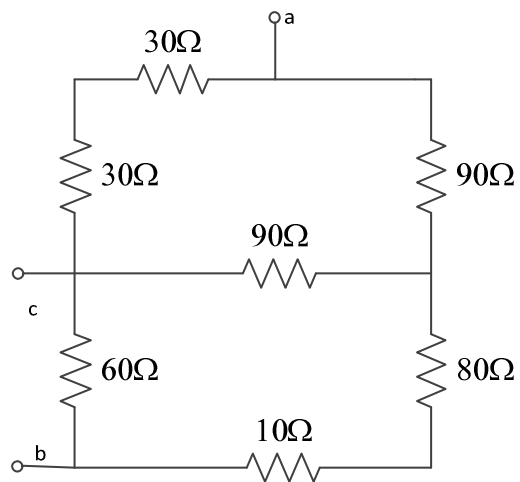
Vir onderstaande kringbaan, bepaal:

- a) R_{cb}
- b) R_{ab}

Question 2.6 [4]

For the circuit below, calculate:

- a) R_{cb}
- b) R_{ab}



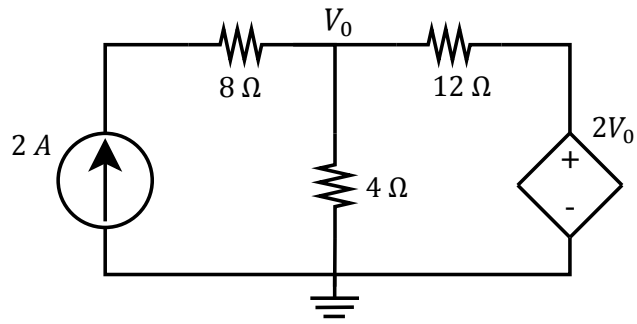
VRAAG/QUESTION 3 [22]

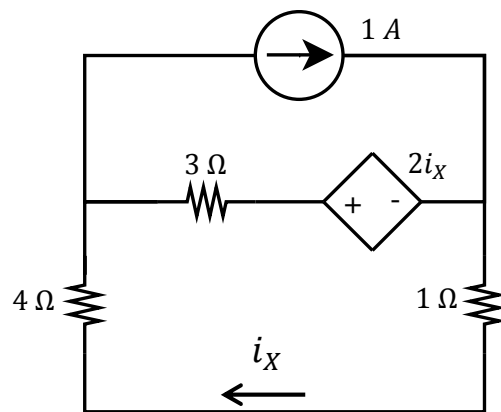
Vraag 3.1 [2]

Gebruik of Node of Lusanalise en bepaal V_0 .

Question 3.1 [2]

Use either Nodal or Mesh analysis to determine V_0 .



Vraag 3.2 [2]Gebruik of Node of Lusanalise om i_X te bepaal.**Question 3.2 [2]**Use either Nodal or Mesh analysis to determine i_X .

Vraag 3.3 [4]

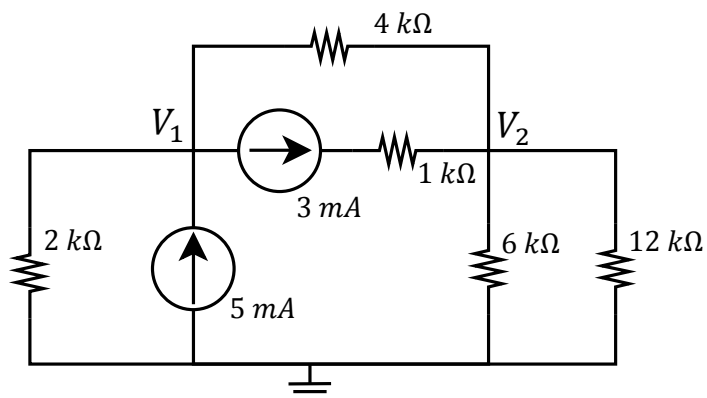
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$\begin{aligned}aV_1 + bV_2 &= 8 \\ cV_1 + dV_2 &= 12\end{aligned}$$

Question 3.3 [4]

Use nodal analysis to set up two equations of the following form:

$$\begin{aligned}aV_1 + bV_2 &= 8 \\ cV_1 + dV_2 &= 12\end{aligned}$$



Vraag 3.4 [4]

Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel:

$$ei_1 + fi_2 = -10 \quad (\text{Mesh equation})$$

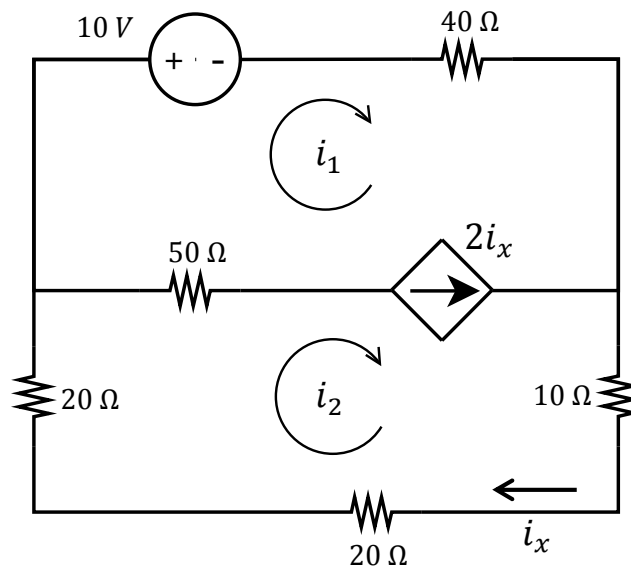
$$i_1 + gi_2 = h \quad (\text{KCL equation})$$

Question 3.4 [4]

Use mesh analysis to set up two equations of the following form:

$$ei_1 + fi_2 = -10 \quad (\text{Mesh equation})$$

$$i_1 + gi_2 = h \quad (\text{KCL equation})$$

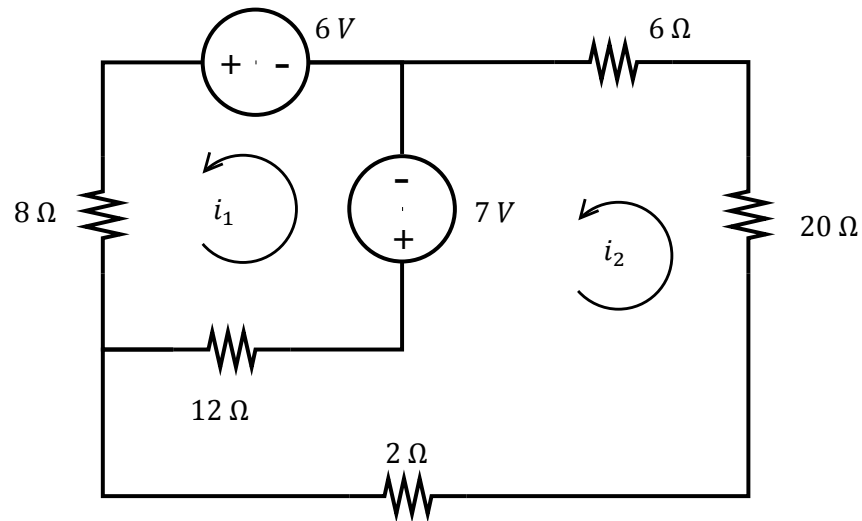


Vraag 3.5 [2]

Stel 'n matriksvergelyking d.m.v. inspeksie op in terme van die lusstrome.

Question 3.5 [2]

Use inspection to set up a matrix equation in terms of the mesh currents.



Vraag 3.6 [4]

Gegee: $\beta = 99$, $V_{BE} = 0.7 \text{ V}$ en $I_B = 5 \mu\text{A}$.

- a) Bepaal V_{CE} .
- b) Bepaal V_S .

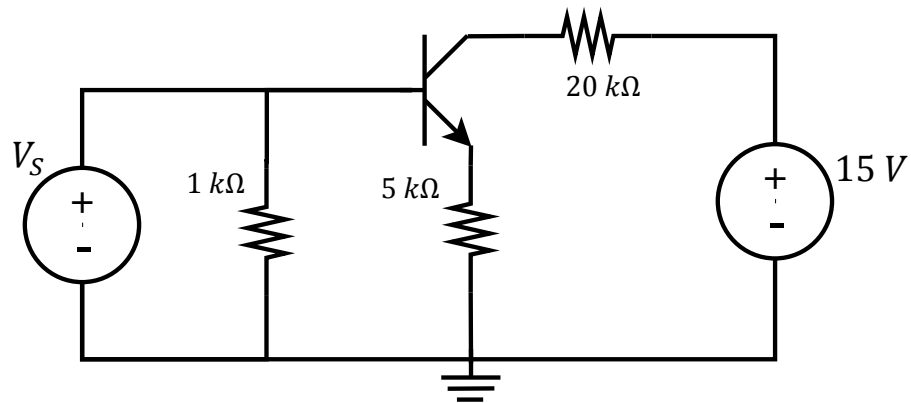
(Vrae a) en b) is onafhanklik van mekaar.)

Question 3.6 [4]

Given: $\beta = 99$, $V_{BE} = 0.7 \text{ V}$ and $I_B = 5 \mu\text{A}$.

- a) Determine V_{CE} .
- b) Determine V_S .

(Questions a) and b) are independent of one another.)



Vraag 3.7 [4]

a) Bepaal k indien $i_1 = 6$ A en $i_2 = 6$ A.

$$\begin{bmatrix} 10 & 1 & k \\ -1 & 1 & 0 \\ 0 & 4 & -2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

b) Bepaal V_2 .

$$\begin{bmatrix} 5 & -4 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

Question 3.7 [4]

a) Determine k if $i_1 = 6$ A and $i_2 = 6$ A.

$$\begin{bmatrix} 10 & 1 & k \\ -1 & 1 & 0 \\ 0 & 4 & -2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

b) Determine V_2 .

$$\begin{bmatrix} 5 & -4 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

EBN111 SEMESTER TOETS/TEST 1 (13 MAR 2014)**ANTWOORDBLAD / ANSWER SHEET**

Van / Surname:		Studentenommer: Student number:							
Voorletters / Initials:		Taal / Language	Afrikaans		English				

Q1		Q2		Q3		Tot:	/45
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VRAAG 1 – QUESTION 1 (7)

1.1	$a) p(Is) =$	$b) W =$	$c)$	$d) Q =$
1.2	$P_1 =$	$P_2 =$	$P_3 =$	

VRAAG 2 – QUESTION 2 (16)

2.1	Onafhanklike lusse: <i>Independent loops:</i>	Takke: <i>Branches:</i>	Nodusse: <i>Nodes:</i>
2.2	$I =$	$v_o =$	$R =$
2.3	$v_y =$		
2.4	$I_l =$	$G_l =$	
2.5	$R_{eq} =$		
2.6	$a) R_{cb} =$	$b) R_{ab} =$	

VRAAG 3 – QUESTION 3 (22)

3.1	$V_0 =$			
3.2	$i_X =$			
3.3	$a =$	$b =$	$c =$	$d =$
3.4	$e =$	$f =$	$g =$	$h =$
3.5	$\begin{bmatrix} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} \end{bmatrix}$			
3.6	$a) V_{CE} =$		$b) V_S =$	
3.7	$a) k =$		$b) V_2 =$	